TE WHARE WĀNANGA O TE ŪPOKO O TE IKA A MĀUI
VICTORIA
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SCHOOL OF ECONOMICS
AND FINANCE

# ECON412 International Economics: Trade

Trimester 2, 2018

# Lecture Week #7: Increasing Returns to Scale, Monopolistic Competition, and Trade

11 September 2018

# Overview

- **Readings:** Feenstra Chapter 5.
- Increasing returns to scale and monopolistic competition
- Derive the gravity equation.

# 1. Introduction

- The neoclassical trade theory, e.g., Ricardian theory and the Heckscher-Olin models, assume that
  - all markets are perfectly competitive;
  - constant returns to scale (e.g., Cobb-Douglas production technology).
- In the neoclassical theory, trade arises because of differences (asymmetries) in technology and factor endowments.
  - The more different the countries, the more they should trade.
  - The more different the countries, the more they should specialize in different goods.

- In reality, we observe that
  - a significant portion of trade is between "similar" countries;
  - these countries tend to trade "similar" goods: a large fraction of trade is intra-industry.
- In addition, in neoclassical trade theory,
  - it is difficult to generate predictions on bilateral trade flows.
  - Some effects of trade liberalization are missed (e.g., within industry reallocation).
  - it is hard to think about the role of firms (e.g., exporting vs non-exporting, multinationals).

- "New" trade theory proposes increasing returns to scale and monopolistic competition as an alternative explanation of international trade.
    - Similar countries specialize in different goods to take advantage of large-scale production.
    - Countries may trade similar goods.

## 2. Monopolistic Competition Model

- Krugman (1979).
- Monopolistic competition: large number of firms, each producing a unique variety of a differentiated product, free entry and exit of varieties.
- There are $N$ product varieties, $N$ is endogenous. A fixed number of consumers, $L$, each receives labor income $w$.
- Labor market is perfectly competitive.

## 2-1. Preference

- Consumer solves the following problem

$$\max_{\{c_1,\cdots,c_N\}} U = \sum_{i=1}^{N} u(c_i),\ \text{s.t.}\ \sum_{i=1}^{N} p_i c_i \leq w.$$

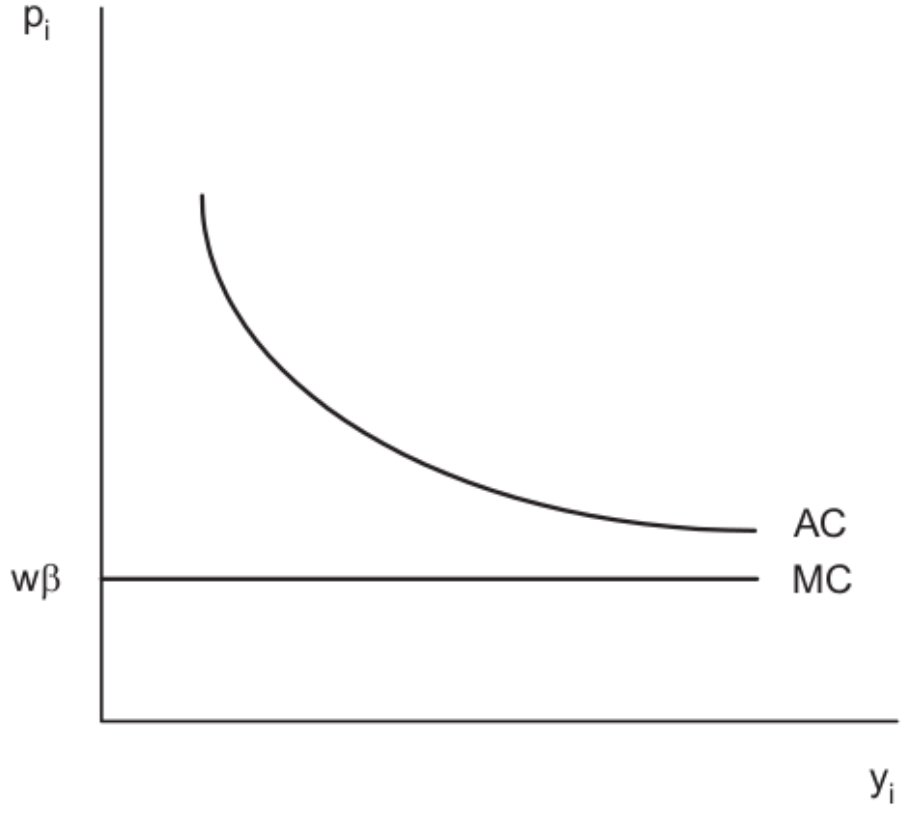
  - **Love of variety**: the larger $N$ is, the higher the utility $U$.
  - Standard assumption: $u' > 0,\ u'' < 0.$
  - Note that $u(\cdot)$ is the same across varieties.

- The first-order necessary condition (FONC) is $u'(c_i) = \lambda p_i,\ i = 1, \cdots, N.$

- Differentiating w.r.t. $p_i$: $u''(c_i)\frac{\partial c_i}{\partial p_i} = \frac{\partial \lambda}{\partial p_i}p_i + \lambda,\ i = 1, \cdots, N.$
  - When $N$ is large, $\frac{\partial \lambda}{\partial p_i}$ is close to zero, suggesting $\frac{\partial c_i}{\partial p_i} = \frac{\lambda}{u''(c_i)} = \frac{u'(c_i)}{u''(c_i)p_i}.$
- Define the price elasticity of demand for variety $i$ as $\sigma_i = -\frac{\partial c_i}{\partial p_i}\frac{p_i}{c_i}$. Then $\sigma_i = -\frac{u'(c_i)}{c_i u''(c_i)} > 0.$
- **Assumption**: $\frac{d\sigma_i}{dc_i} < 0$, elasticity is higher at a lower consumption $c_i$.
  - implying that, at a low consumption, consumption $c_i$ is more elastic to changes in $p_i$.

## 2-2. Production

- Labor is the only production factor, labor requirement of producing $y_i$ unit of variety $i$ is $L_i = \alpha + \beta y_i$.
  - $\alpha$ is the fixed labor requirement. Think about $\alpha$ as a fixed cost to pay in order to produce $i$.
  - $\beta$ is the marginal labor input.
- **Production costs**. Average cost $AC_i = \frac{wL_i}{y_i} = \frac{w\alpha}{y_i} + w\beta$; Marginal cost $MC_i = w\beta$.


$p_i$
AC
$w\beta$
MC
$y_i$

## 2-3. Monopolistic competition

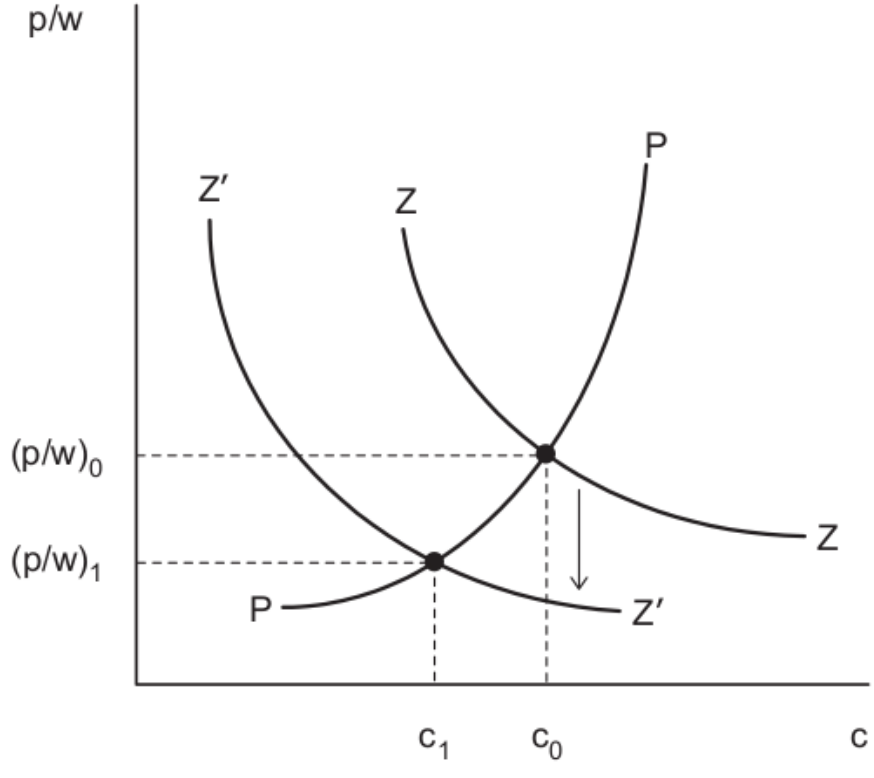
- Key equilibrium conditions:
  - Each firm maximizes its own profits by setting marginal revenue equal to marginal cost ($MR = MC$).
  - Free entry when profits are positive, implying zero profits (in the long-run equilibrium), $p = AC$, average revenue=average cost.
- The underlying problem is "symmetric", i.e., prices, as well as quantities, will be the same across varieties in equilibrium.

- The **profit maximization** (optimal pricing) for each variety is given by (omitting subscript $i$):
  - $MR = MC$: $p\left(1 - \frac{1}{\sigma}\right) = w\beta$, or $\frac{p}{w} = \beta\frac{\sigma}{\sigma-1}$.
- To see this, first notice that the firm supplies $y_i$ to meet the demand $L \cdot c(p, w)$, and solves the following
$$\max_p py - w[\alpha + \beta y].$$
FONC: $y + p\frac{dy}{dp} - w\beta\frac{dy}{dp} = 0$, which can be written as $(p - w\beta)\frac{dy}{dp} = -y$. Multiply both sides by $-\frac{p}{y}$, we get $(p - w\beta)\left(-\frac{dy}{dp}\frac{p}{y}\right) = p$. Note that $-\frac{dy}{dp}\frac{p}{y} = \sigma$, the demand elasticity (of consumers). We then have $p\left(1 - \frac{1}{\sigma}\right) = w\beta$, which can be also written in the familiar form.
$$p = \frac{\sigma}{\sigma - 1}w\beta.$$
- The **free-entry** (zero-profit condition) means that $py - w\beta y = w\alpha$. This gives
  - $p = \frac{w\alpha}{y} + w\beta$, average revenue = average cost.

## 2-4. Equilibrium

- Output markets clear: $y_i = Lc_i$. Symmetry implies we can omit subscript $i$.
- We can solve the two equilibrium conditions, for $\frac{p}{w}$ and $c$.
  - Only relative price $p/w$ can be solved: doubling price and wage at the same time does not change result (see budget constraint).
  - Under the assumption $\frac{d\sigma}{dc} < 0$, price and consumption are obtained crossing of PP and ZZ.
  - PP: $\frac{p}{w} = \frac{\sigma}{\sigma-1}\beta$, profit maximization.
  - ZZ: $\frac{p}{w} = \frac{\alpha}{Lc} + \beta$, free-entry profit.
- labor market clears: $L = \sum_{i=1}^{N}(\alpha + \beta y_i) = N(\alpha + \beta Lc)$. This leads to the equilibrium number of varieties, as follows

$$N = \frac{L}{\alpha + \beta Lc}.$$


p/w
P
Z′
Z
(p/w)0
(p/w)1
Z
P
Z′
c1
c0
c

## 2-5. International trade

- Suppose that two countries have identical size ($L = L^*$). In the Heckscher-Ohlin model, they do not trade.
- In the monopolistic competition model, the two countries trade because firms produce differentiated products and consumer's preference displays the love of variety.
- The increased number of competitors will lower the output prices.

- Under free trade, labor endowment doubles, the $ZZ$ curve shifts down to $ZZ'$, price $p/w$ falls, real wage increases.
  - **Individual consumption** of each variety also decreases (spreading expenditure over more varieties).
  - Total number of variety increases, $N^T = \frac{1}{\alpha/(2L)+\beta c^T}$. $c^T$ is smaller than autarky.

- Output of each variety becomes larger: $p/w$ is now lower, $y$ will be larger due to higher demand elasticity (at the lower level of individual consumption). In fact, from the free-entry condition, we also know that $y$ rises because $p/w$ falls.
- Each country now produces a smaller number of variety, $N = \frac{L}{\alpha+\beta y}$, which is smaller as $y$ becomes larger, relative to the autarky.
    - Some firms exit following trade.
- Gains from trade: 1) higher real wage; 2) more variety.

- Impacts of trade on productivity
  - *Scale effect*: surviving firms expand their production, because a larger demand elasticity at the lower individual consumption. Fixed cost implies that a higher output means a higher output per unit of labor.
  - *Selection effect*: Implicitly, some firms cannot survive. This is rather a derived implication, than the model's explicit result. In model,varieties are symmetric.
- Empirical evidence suggestst that the selection effect dominates.
  - Pavcnik (2002) and Trefler (2004).

## 3. Monopolistic Competition with CES Preference

- Two effects of increased trade in the above model can be estimated in empirical tests.
- Some empirical results suggest that the *scale effect* is small, while the gains from trade are due to the *selection effect*.
  - Trefler (2004) in the case of Canada-U.S. trade agreement; Pavcnik (2002) in the case of Chile.
- This suggests that the elasticity of demand for product varieties may be *constant*., i.e., $\frac{d\sigma_i}{dc_i} = 0$.

## 3-1. CES preference and monopolistic competition

- Krugman (1980)
- A utility function that leads to constant demand elasticity is one exhibiting **constant elasticity of substitution (CES)**, as follows

$$U = \sum_{i=1}^{N} c_i^{\frac{\sigma-1}{\sigma}}$$

where $\sigma > 1$ is the (constant) elasticity of substitution between any pair of varieties.

  - What is the elasticity of substitution? $\sigma = \frac{d\ln(c_i/c_j)}{d\ln|\partial c_i/\partial c_j|}$.
  - $\frac{\partial c_i}{\partial c_j}$ is the marginal rate of substitution between $i$ and $j$, the slope of the indifference curve.
  - $\sigma$ is the curvature of indifference curve.

- Under the CES preference, markup is $\frac{p}{w} = \frac{\sigma}{\sigma-1}\beta$ for each variety, which is constant.
- From the free-entry condition, the output of firms is given by $\frac{p}{w} = \frac{\alpha}{y} + \beta$. Or, $\bar{y} = (\sigma - 1)\alpha/\beta$, also constant.
- The number of varieties *produced* is also constant, $N = \frac{L}{\alpha+\beta\bar{y}}$, or $N = \frac{L}{\alpha\sigma}$.

- **Gains from trade**: Suppose that two countries have the same preferences and technologies, but differ in labor endowments.
- Consumers in each country now consume $N + N^*$ varieties.
  - Consumers in home country spend a fraction $N^*/(N + N^*)$ of their income on foreign goods.
- The value of home country's imports measured in wage units is $L\frac{N^*}{N+N^*} = L\frac{L^*}{L+L^*}$.
- The value of trade is determinate, but the *direction* of trade is not.
- Gains from trade arise from the increased number of varieties.
  - There are no longer the scale effect and the selection effect of international trade.

## 3-2. Implication of trade cost: the home market effect

- **Home-market effect**: countries tend to export goods for which they have relatively large markets.
- Trade costs are mostly absent from the neoclassical theory of international trade.
- We extend the model in 3-1, to explore the implications of trade cost in the presence of increasing returns to scale (IRTS).
    - Factor Price Equalization (FPE) no longer holds in the presence of trade cost.
- Main idea: Because of IRTS, firms choose to have one location only. Because of trade costs, firms prefer to locate in large markets.

- Two countries: Home and Foreign. They are **identical** except labor endowments: $L^H, L^F$.
- CES preference: $U = \sum_{i=1}^{N} (c_i^H)^{\frac{\sigma-1}{\sigma}}$ at Home. Similar for Foreign.
- Production technology is as before: producing $y_i$ units of $i$ requires $L_i = \alpha + \beta y_i$ units of labor.
- Iceberg trade cost: To deliver 1 unit of a good to another country, the exporting country must produce and ship $\tau > 1$ units of the good.

**Consumer's utility maximization**

- CES preference is homothetic, the optimal consumption of variety $i$ as a share of income is constant.
- At Home, individual's optimal consumption of variety $i$ is $c_i^H = \left(\frac{p_i^H}{P^H}\right)^{-\sigma} \frac{w^H}{P^H}$
  - where $P^H = [\sum_{j=1}^{N} (p_i^H)^{1-\sigma}]^{\frac{1}{1-\sigma}}$, the aggregate price index.
- Symmetry: in equilibrium, firms' optimal prices are the same, optimal consumption is the same, across varieties *produced within the country*.
- Then, in Home, $P^H = [N^H (p^H)^{1-\sigma} + N^F (\tilde{p}^F)^{1-\sigma}]^{\frac{1}{1-\sigma}}$. In Foreign, $P^F = [N^H (\tilde{p}^H)^{1-\sigma} + N^F (p^F)^{1-\sigma}]^{\frac{1}{1-\sigma}}$.
- **Optimized consumption** at Home: domestic goods $C^{H,H} = \left(\frac{p^H}{P^H}\right)^{-\sigma} \frac{w^H}{P^H} L^H$, and imported goods $C^{F,H} = \left(\frac{\tilde{p}^F}{P^H}\right)^{-\sigma} \frac{w^H}{P^H} L^H$. Similarly in Foreign.

**Firm's profit maximization**

- **Monopoly pricing**: $p^H = \frac{\sigma}{\sigma-1}\beta w^H$, and $p^F = \frac{\sigma}{\sigma-1}\beta w^F$.
- Export prices: $\tilde{p}^H = \tau p^H$; and $\tilde{p}^F = \tau p^F$.
- Profit of a firm at Home is
  $\pi^H = (p^H - \beta w^H)C^{H,H} + (\tilde{p}^H - \beta w^H)C^{H,F} - \alpha w^H$.
- **Free entry** implies zero profit:
  $(p^H - \beta w^H)C^{H,H} + (\tilde{p}^H - \beta w^H)C^{H,F} = \alpha w^H$ at Home. Similarly in Foreign.

**Equilibrium**

- **Labor market clearing** at Home: $L^H = N^H[\alpha + \beta(C^{H,H} + C^{H,F})]$.
- As derived earlier, the number of varieties: $N^H = \frac{L^H}{\alpha\sigma}$ at Home, $N^F = \frac{L^F}{\alpha\sigma}$ in Foreign.
- Balanced trade condition: $N^H \tau p^H C^{H,F} = N^F \tau p^F C^{F,H}$.

**Market size and wage**

- Substitute for $N^H$, $N^F$, $C^{H,F}$, and $C^{F,H}$ in the balanced trade condition.
- We obtain the relative wage:

$$(w^H/w^F)^\sigma = \frac{\tau^{1-\sigma} + (L^H/L^F)(w^H/w^F)^{1-\sigma}}{1 + \tau^{1-\sigma}(L^H/L^F)(w^H/w^F)^{1-\sigma}}.$$

- **Proposition**: $w^H \geq w^F$ if and only if $L^H \geq L^F$.
  - A larger market has a higher wage.
- **Intuition:**
  - Everything else equal, demand is larger in larger markets.
  - Firms will only be active in the smaller market if they can pay lower wages or face less competition (i.e., smaller number of domestic firms).
  - Here, labor supply is *perfectly inelastic*: number of domestic firms is fixed.
  - So the smaller market has to be associated with lower wages.

**Home market effect**

- The above model does not predict the following evidence:
  - Larger countries produce disproportionately more varieties.
- To generate the home market effect, we need another numeraire homogeneous good which can be freely traded, we assume that it is produced one-for-one with labor.
- Further assume the preference over the differentiated good and the numeraire is Cobb-Douglas.
- This suggests that the wage will equalize between Home and Foreign: $w^H = w^F$.

- **Proposition** (home-market effect): $N^H/L^H > N^F/L^F$ if and only if $L^H > L^F$.
- Intuition: given the equal wage across countries, firms will be active in a smaller market only if competition is softer (fewer varieties).
- It can be shown that
$$N^H/N^F = \frac{L^H/L^F - \tau^{1-\sigma}}{1 - \tau^{1-\sigma}(L^H/L^F)}.$$
- Proposition can be shown from this equation.